Gabriel Eze

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Education

B.Sc. Data Science & Mathematics, *Centre College* Danville, USA | *08/2022 – 05/2026*

Relevant Coursework: *Abstract Algebra, Advanced Interpolation Methods, Applied Machine Learning, Bayesian Data Analysis, Building AI-Powered Applications, Data Manipulation & Visualization, Linear Algebra, Mathematical Statistics, Probability Theory, Real Analysis, Statistical Modeling, Topics in Linear Algebra*

Research

Stack Domination Density of Hypercubes, *Graph Theory* Dr. Brauch, Dr. Wiglesworth | *05/2025 – 08/2025*

- Formulated and proved results for stack domination density in $\boldsymbol{C}m \times \boldsymbol{P}2 \times \boldsymbol{P}n$, deriving bounds across residue classes of m.
- Designed modular "Clockwise" and "Domination" algorithms to construct minimal dominating sets, integrating combinatorial proofs and computational verification into a 38-page manuscript for potential undergraduate publication.

Pitch Modeling , *Carnegie Mellon SURE Camp* Quang Nguyen, Dr. Yurko | *06/2024 – 07/2024*

- Applied Random Forest, Lasso regression, and HGBR models to predict pitch effectiveness for MLB players, leveraging large datasets from FanGraphs and Baseball Savant.
- Developed a pitch recommendation tool to suggest pitch sequences.

In-person presentation @ CMSAC 2024—sole on-site team

Sports Analytics Dr. Heath, Dr. Poudel | *05/2023 – 07/2023*

- Utilized Python to lead data cleaning and parsing efforts within a complex big data pipeline, ensuring data accuracy and reliability for assessing player substitutions.
- Collaborated on the development of win probability models.

Elevator pitch + poster presentation @ CMSAC 2023

Personal Projects

Family Business Bookkeeping , *Archkings Confectionery Ltd* *09/2026 – Present*
Tracking income, expenditure, and operating costs.

Forest Fire Simulation *11/2022 – 12/2022*
Designed class-based forest simulation modeling state transitions on a 10×15 grid.

DNA Sequence Analysis *10/2022 – 11/2022*
Developed sliding-window STR detection algorithm to quantify repeat sequence from genomic input.

Skills

- **Computing** — Q-Ball Imaging
- **LLM** — Chunking, Doc2Query, Embeddings, Hit@K, Prompt Engineering, Red-Team Testing, Tool Use
- **Programming** — MATLAB, Python, R, SQL
- **Tools** — ChromaDB, Endpoints, Git, JupyterLab, LaTeX, MySQL, Pydantic, Quarto, Regex, Tableau, VS Code

Awards

Best Visualization , *ASA DataFest @ EKU* *2025*
Analyzed national office leasing data to deliver strategic market insights for Savills in 3-day competition.